

Assignment #2 – Motor speed control
ME 360 – Electromechanical Design
Spring 2025, release date: 01/29/2025

This assignment has individual and team components and will take 3 weeks to complete. It is due in class on 03/05 (Sections A1, A2) and 03/06 (Sections A3, A4), and as an entry in your design portfolio on 03/21.

The goal of the assignment is to apply the concepts and methods of motor speed control that we have reviewed in class to the simulation and design of a platform that transports a dynamically unstable load.

The learning objectives are:

- To use advanced computational tools in the design process, to validate ideas and design concepts
- To implement a control strategy to maintain constant rotor speed
- To develop physical prototyping skills for the validation of mechanical designs

This exercise will introduce you to multibody dynamic simulation, which is a widely used technique for the design and improvement of mechanisms and machines, and their components.

The goal of this exercise is very simple: design a system that transports a 1-foot long vertical prismatic bar across a variable distance, from 5-10 ft, and back, stop to stop, without toppling. The bar is to be placed on a horizontal platform, and will be supported at its bottom face only. No glue, fasteners or side supports are allowed.

The objective is to minimize the time required to transport the bar.

DELIVERABLE 1 (Individual) – Multibody dynamic model (Motion simulation with rigid components) = 35%

You will develop one or more design concepts to perform the required motion, and determine the minimum time required using multibody simulation in Solidworks. It is recommended that you derive the stability conditions of the system as a point of comparison to ensure that your simulation yields sensible results.

You will upload sketches of your design ideas and at least one full simulation ZIP file consisting of CAD parts and assembly, with the motion study that illustrates the conditions under which you achieved the minimum time for the motion. Create an entry for this project in your design portfolio and clearly state your record time. Your results may differ from those of your peers, but you must explain why you think yours make sense.

DELIVERABLE 2 (Same teams formed for assignment 2) – Class competition = 65%

You will create a physical prototype that validates an efficient control strategy. Teams will compete to achieve the motion of the bar, without toppling, in the shortest time possible. The bar shall stand at rest at the end of the test. Time is stopped when motion of the bar has completely subsided. The distance to travel will be selected at random before your test, so make sure you have a personal computer available to upload the updated sketch onto the Arduino.

The trials will take place during class on the last day of the project (see “Timeline for the Assignment” below).

You will create an entry in your design portfolio that describes your design process, decisions, construction, tests, iterations and your competition time (video).

Grading will be based on the effectiveness of the design (does it achieve the goal?), its efficiency (how fast is it?), craftsmanship (are the parts and assembly carefully designed and joined, attention to detail?) and teamwork (fair distribution of tasks).

The awards for the winners of the competition will consist of stunning ME merchandise items (TBD).

Peer evaluation

Towards the end of the project you will be asked to fill out a peer evaluation form in which you will describe the tasks that each team member performed and the individual degree of participation, based on your judgement. This feedback will be used to assign credit for the project fairly among the team members. Team members with low participation scores will have the opportunity to make up for lost credit in a final evaluation that will take place in the last day of classes.

Supplementary information

The bar will be 1 foot long, made of 8020 aluminum extrusion. For more information and the exact geometry, checkout this link: <https://8020.net/1010.html>

You will be provided with a 12V gear motor, an Arduino UNO board with power amplifier (red board), a 12V adapter and a USB cable.

You will use foam board and 3D printed parts to build your physical prototype. You may use any materials such as fasteners and pins, that the EPIC is able to provide.

You will have access to a variety of O-rings that you can use to design wheels of different sizes. Ask the instructor during your team’s meeting times.

Timeline for the assignment

02-03/02-11: Closed-loop motor control

02/12-03/04: Team meetings

03/05-03/06: Prototype testing and trials

03/21: Final entry in your design portfolio